Learned Z-Score Forecasting for Cross-Exchange Cryptocurrency Spreads: A Gradient-Boosting Approach on Live Microstructure Data

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Abstract

Identical cryptocurrencies often trade at different prices across exchanges. Most high-frequency pairs strategies still turn that gap into a trade with a simple rule: enter when the rolling z-score of the spread is large, and wait for mean reversion. We instead *forecast* the next z-score and show that a *tree-based* ML model can outperform those mechanical z-score rules under live cross-exchange trading. We study *same-asset cross-exchange* spreads at roughly one-minute cadence, using live multi-venue L2 and related microstructure features that historical candle APIs cannot rebuild. Unlike the DNN/LSTM architectures common in prior spread forecasting [4, 7, 10–12, 15], we use a tree-based ML model on tabular features (lags, coin/venue categoricals, and live microstructure) that is strong for this data shape and still uncommon for our problem setting. LightGBM forecasts the next-snapshot cross-exchange z-score z_{t+1}; we also compare against a classical Random Forest baseline as a control on the same out-of-sample test set. We systematically select a confidence filter on that forecast, trading only when $|\hat{z}_{t+1}| \geq 0.9$: the least restrictive threshold at which per-trade Sharpe reaches ≥1. On the live validation book under this filter, LightGBM achieves about 87% directional accuracy and R^2 of 0.599 across 12,795 closes, with an hourly portfolio Sharpe of 1.90. Under matched capacity at the same gate, directional accuracy improves by 15.4 percentage points over mechanical z-score persistence. On the test set, Random Forest tracks LightGBM's ranking closely, while LightGBM remains preferred for more trades at the gate and higher R^2. We publicly release the code repository and trained model on GitHub (the code repository (URL withheld for double-blind review)) and the synchronized multi-exchange microstructure dataset on Hugging Face (the public dataset release (URL withheld for double-blind review)) so others can reproduce and extend the work.

CCS Concepts

• **Computing methodologies** → **Supervised learning by regression**; • **Applied computing** → *Economics*; • **Information systems** → *Data mining*.

Keywords

statistical arbitrage, cryptocurrency, cross-exchange spreads, gradient boosting, market microstructure, limit order book, live evaluation

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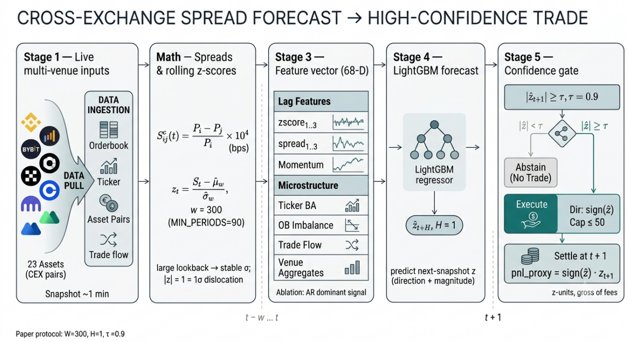

Figure 1: End-to-end paper protocol: live multi-venue inputs → cross-exchange spread and rolling z-score (w=300) → 68-D features (AR lags dominant) → LightGBM forecast $\hat{z}_{t+1}$ (H=1) → confidence gate $|\hat{z}| \geq \tau$=0.9 → selective trade and t+1 settlement in z-units.

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1 Introduction

Cryptocurrency markets are fragmented across dozens of centralized exchanges (CEXs), each with independent order books, heterogeneous fees, liquidity, and matching-engine latencies. Unlike equities, there is no consolidated tape that forces immediate price convergence. The result is recurrent cross-exchange price dislocations for identical assets [8]. These gaps present economically meaningful opportunities for same-asset statistical arbitrage, especially among high-volatility coins where short-lived Law-of-One-Price deviations appear at approximately one-minute granularity.

Existing high-frequency crypto pairs-trading systems largely treat the rolling z-score of a spread as a mechanical entry/exit rule: trade when $|z_t|$ exceeds a fixed threshold and exit on reversion [2, 6, 9, 13]. However, this design is not as reliable in the cross-exchange setting. At minute scale, spreads sometimes revert quickly and sometimes persist through funding or liquidity regimes; a fixed threshold always bets on mean reversion without understanding the broader context of the exchange dynamics at that moment. Venues also differ systematically in fees, depth, and latency, so a global rule treats every (coin, exchange-pair) identically, while missing potential systematic differences between exchanges. Checking z_t at every snapshot is cheap, but it leaves these two sources of structure unused, which may in turn impact the quality of both the direction and magnitude of the short-horizon forecasts.

Here we replace thresholding the current z-score with a supervised forecast of the next one. Rather than adopting the DNN/LSTM architectures common in prior spread-forecasting work, we train a single LightGBM [5] panel model—a simple tabular tree baseline suited to heterogeneous lag and microstructure features with native categoricals—to predict $y_t = z_{t+H}$ (with H=1 in the production protocol) for same-asset cross-exchange spreads, conditioning on lagged spread dynamics, live L2 orderbook imbalance and trade flow, and native `coin` / venue-pair categoricals, as well as data about the trades, funding rate, and open interest. Bid/ask imbalance, which comes from analysing the L2 order book, is especially of interest to us in this study, as it encodes directional pressures and slippage risk that mechanical z-score rules can never access, and it cannot be reconstructed from historical OHLCV APIs. A confidence filter is implemented which then trades only when $|\hat{z}| \geq 0.9$, selecting high-magnitude z-score forecasts to trade on, rather than forcing an action on every snapshot, even when there is no arbitrage opportunity.

Methodologically, this is a change of formulation, not only of market setting. The prediction target is a local mean–standard-deviation normalization suited to short-horizon directional and magnitude learning; the model architecture learns venue- and asset-specific dynamics that global thresholds cannot represent; and finally the live-collected microstructure features define an evaluation setting that backtest-only studies cannot replicate. We enforce a chronological train/test split, score capacity-matched mechanical baselines on identical rows and settlement rules, and validate end-to-end in live paper-trading under the system's real collection and decision latency, including multi-day campaigns at approximately one-minute granularity. We report on our model's directional accuracy, R^2, and `pnl_proxy`. The synchronized multi-exchange signal stream is released publicly through the public dataset release (URL withheld for double-blind review)as a standalone dataset contribution.

1.1 Contributions

(1) **Learned next-snapshot z-score forecasting for same-asset cross-exchange spreads.** We replace mechanical $|z_t|$ thresholds with a tabular tree forecast of z_{t+H} (LightGBM [5]; Random Forest [1] as a classical control baseline on the test set), conditioned on live microstructure and coin/venue categoricals, with a $|\hat{z}| \geq 0.9$ confidence gate for selective entry.

(2) **Live paper-trading under real decision latency.** We run the policy in live campaigns (including a ~72-hour session) on direct centralized-exchange market data at approximately one-minute frequency: the trained LightGBM model predicts the next z-score at each snapshot and executes paper trades under the system's real collection and decision delay.

(3) **Clear gains over matched mechanical baselines under live validation.** Scoring in z-units (not dollars), the paper gate $|\hat{z}| \geq 0.9$ on the ≈72-hour validation book yields about 87% directional wins and mean `pnl_proxy` of +1.37 on 12,795 closes; under matched capacity at τ=0.9, the learned policy improves win rate by 15.4 percentage points and mean `pnl_proxy` by 63% over capacity-matched z-score persistence under identical settlement rules.

(4) **Public code, model, and microstructure dataset.** We release the repository, trained LightGBM model, and synchronized multi-exchange signals across 23 volatile assets and 6 exchanges (the code repository (URL withheld for double-blind review); the public dataset release (URL withheld for double-blind review)), including L2 orderbook and trade flow that cannot be reconstructed from standard historical exchange APIs, plus funding and open interest for downstream use.

2 Related Work

Classical cryptocurrency pairs trading largely focuses on methods used in statistical arbitrage and pairs trading in equity markets. They apply distance, cointegration, and z-score rules to single-exchange, different-asset pairs. Fil and Kristoufek [2] implement distance and cointegration methods to a crypto pairs trading setting, and find that they underperform classical benchmarks, while showing that performance is highly sensitive to parameters such as the collection frequency, and the execution window. Other studies present rule-based trading frameworks for crypto markets, [6, 9, 13, 14], however spread persistence between coins and across exchanges may vary with liquidity and funding regimes, venues differ in fees and latency, and high-volatility assets can produce frequent but short-lived price dislocations. Fixed thresholds cannot distinguish fast from slow reversion episodes, and they treat every venue pair identically. Therefore, we decided to explore same-asset cross-exchange opportunities, where the Law of One Price helps guarantee mean reversion, using a learned model, which can better exploit the different patterns and asymmetries present in the volatile crypto market. The target variable in our model is the forecast of the future rolling z-score, which informs whether or not we should trade on a certain asset.

A smaller literature trains models on spread or relative-return dynamics. Fischer et al. [3] show that machine-learning statistical arbitrage is feasible at minute scale in cryptocurrency markets, motivating our focus on a tighter same-asset cross-exchange target rather than cross-sectional coin ranks from lagged returns. They also study sensitivity to execution delay; they find that execution on a profitable trade needs to occur within less than five minutes from the detected arbitrage opportunity, or else the opportunity for profits disappears entirely. This finding helps motivate our decision to forecast the next z-score, as live trading entails execution delays from the time a transaction is initialized to when it is settled. Therefore, forecasting the next z-score and trading based on these predictions helps circumvent this practical challenge. We then take a further step by deploying the model in live paper-trading campaigns on real-time collected signals, so decisions are made under the system's actual collection and decision latency and scored against subsequent market outcomes. Closely related supervised work often adopts *sequence or deep* architectures—DNN/LSTM crypto spread models, equities LOB predictors, and abstention-based filters [4, 7, 10–12, 15]. Those inductive biases are natural for long raw histories, but they are a different modeling choice from ours. We instead treat the problem as a *tabular* panel of engineered lags and live microstructure features (including native `coin` / venue-pair categoricals) and use *tree ensembles* as a simple baseline class for minute-scale next-z forecasting. Direct numeric comparisons to published DNN/LSTM results remain imperfect because none of that prior work combines minute-scale same-asset cross-exchange pairs trading under live multi-venue L2 collection; our differentiation is therefore both the setting and the intentional tree-based formulation, not a deferred bake-off against those architectures.

Methodologically, the contribution is a simple confidence-gated tree baseline for that setting rather than a novel deep architecture. The supervised prediction target is the future rolling z-score $y_t = z_{t+H}$ (local mean–standard-deviation normalization for short-horizon direction and magnitude); a confidence filter then trades only when $|\hat{z}| \geq 0.9$. We train a LightGBM [5] gradient-boosted tree on coin and venue-pair categoricals, ticker, orderbook, trade, funding, and open interest features collected in real time—signals that inform short-horizon movement and cannot be reconstructed from historical OHLCV APIs (released with the public dataset release (URL withheld for double-blind review)). A classical Random Forest [1] scored on the same train/test feature matrix serves as a control baseline within the tree family (Section 5); capacity-matched mechanical baselines that threshold the current z-score at the same gate remain the primary trading peers under live validation. We evaluate directional accuracy and R^2 for forecast quality and `pnl_proxy` plus hourly portfolio Sharpe for trading stability. Live campaigns, including a multi-day ~72-hour validation session at approximately one-minute granularity, test whether test-set skill holds on real-time, non-replayable market state.

3 Methodology

3.1 Cross-Exchange Spread and Z-Score Target

Let $P_i^c(t)$ denote the mid-price of asset c on exchange i at snapshot t. For an exchange pair (i, j), the cross-exchange spread in basis points is:

$$S_{ij}^c(t) = \frac{P_i^c(t) - P_j^c(t)}{P_i^c(t)} \times 10{,}000 \tag{1}$$

Unlike traditional pairs trading where the spread between different assets may diverge permanently as fundamentals change, the cross-exchange spread for the same asset is anchored by the Law of One Price: absent frictions, $S_{ij}^c(t) \to 0$. In practice, fees, withdrawal delays, and matching-engine latency allow temporary price deviations for the same coin across different exchanges.

We normalize the spread into a rolling z-score:

$$z_t = \frac{S_t - \hat{\mu}_w}{\hat{\sigma}_w} \tag{2}$$

where $\hat{\mu}_w$ and $\hat{\sigma}_w$ are the estimated rolling mean and standard deviation over a window of w snapshots, with a minimum period requirement before the z-score is defined. Model outputs $\hat{z}$ are therefore in units of this trailing $\hat{\sigma}_w$: $|\hat{z}| = 1$ is a one-σ forecasted dislocation. We use $w = 300$ (MIN_PERIODS $= 90$); larger w adds context but lengthens warmup (Section 6).

3.2 Supervised Prediction Target

The model's target is the *future* z-score at horizon H:

$$y_t = z_{t+H} \tag{3}$$

computed via `groupby(coin, pair).zscore.shift(-H)` on the training panel. This is a regression target, not a binary classification. The model predicts both the direction and magnitude of the z-score H snapshots ahead, enabling a confidence filter on $|\hat{z}|$.

For the prediction model used in the paper-trading campaigns, $H = 1$ (predict the next snapshot's z-score), $w = 300$ snapshots, and MIN_PERIODS $= 90$. Window length and confidence cutoff (τ) selection is described in Section 6.

3.3 Feature Engineering

The feature matrix is constructed from four synchronized signal families collected at each snapshot: spread matrices, ticker quotes, L2 orderbook, and recent trades. Features are computed per (coin, exchange-pair, snapshot) observation.

3.3.1 Spread and z-score features. Lagged values of the spread in basis points and the rolling z-score form the base feature set:

- `spread_bps_lag`$\{1, 2, 3\}$: raw spread at lags 1–3.
- `zscore_lag`$\{1, 2, 3\}$: rolling z-score at lags 1–3.
- `spread_momentum_1_3`: difference between lag-1 and lag-3 spread, capturing short-term trend.
- `zscore_momentum_1_3`: same for z-scores.
- `zscore_accel`: second-order difference $z_{t-1} - 2z_{t-2} + z_{t-3}$, capturing acceleration in z-score trajectory.

3.3.2 Ticker microstructure features. From each exchange's best-bid/ask quotes:

- `tk_mid_lag`$\{1, 2\}$: lagged mid-prices per exchange.
- `tk_spread_bps_lag`$\{1, 2\}$: lagged bid-ask spreads in bps per exchange. Wider spreads indicate lower liquidity and higher execution cost.
- `tk_bid_volume_lag`, `tk_ask_volume_lag`: lagged top-of-book volumes per exchange.

3.3.3 Orderbook imbalance features. From the top 20 price levels on each side of each exchange's order book, we compute:

$$\texttt{imbalance}_i = \frac{V_{\text{bid},i} - V_{\text{ask},i}}{V_{\text{bid},i} + V_{\text{ask},i}} \tag{4}$$

where $V_{\text{bid},i}$ and $V_{\text{ask},i}$ are the total volumes on the bid and ask sides of exchange i's top-20 levels. This ratio ranges from -1 (entirely ask-dominated) to $+1$ (entirely bid-dominated) and captures directional pressure in the order book. Lagged imbalance values `ob_imbalance_lag`$\{1, 2, 3\}$ for each exchange provide the model with recent order flow dynamics.

3.3.4 Trade flow features. From the most recent trades on each exchange:

- `tr_buy_sell_ratio`: ratio of buy-initiated to sell-initiated trade volume, indicating whether recent execution flow is dominated by buyers or sellers.
- `tr_total_volume`: total trade volume, a proxy for activity and liquidity.

Both are lagged at 1–2 snapshots per exchange.

3.3.5 Cross-exchange aggregates. Features that capture the state of the multi-venue market as a whole:

- `cross_mid_range`, `cross_mid_std`: range and standard deviation of mid-prices across exchanges, measuring instantaneous cross-venue dispersion.
- `cross_ba_std`, `cross_ob_std`, `cross_ob_range`: cross-exchange variability in bid-ask spreads and orderbook imbalance.
- `net_ob_pressure`: aggregate orderbook imbalance across venues.
- `price_rank_{exchange}`: ordinal rank of each exchange's mid-price within the venue set, identifying which exchange is currently pricing high or low.
- `tightest_ba`, `widest_ba`, `ba_range`: spread-liquidity extremes across venues.

3.3.6 Categorical identifiers. `coin` and `pair` (exchange-pair label) are encoded as LightGBM native categoricals, allowing the model to learn asset-specific and venue-pair-specific dynamics without separate per-pair models.

The prediction model uses 68 features in total.

3.4 Model: LightGBM Gradient Boosting

We train a single LightGBM [5] regression model on the pooled panel of all (coin, exchange-pair, snapshot) observations. The objective is squared error minimization on $y_t = z_{t+H}$.

We deliberately choose tree ensembles over the DNN/LSTM architectures common in prior spread-forecasting work: the feature matrix is a heterogeneous tabular panel (lags, microstructure, and native categoricals) for which tree models are a strong default inductive bias. As a classical Random Forest baseline on the same train/test feature matrix, we also fit a Random Forest regressor [1] (regularized leaf size analogous to LightGBM's minimum child samples). LightGBM is the model used for validation-campaign evaluation; the Random Forest control is scored on the test set (Section 5).

3.5 Trading Policy

At each live snapshot, the model generates a prediction $\hat{z}_{t+1}$ for every (coin, exchange-pair) observation. The trading policy is:

(1) Entry: open a position if $|\hat{z}_{t+1}| \geq \tau$ and the (coin, pair) slot is not already occupied. The paper protocol uses τ=0.9.
(2) Direction: $\text{sign}(\hat{z}_{t+1})$. The model predicts the sign of the spread z-score at $t+1$ and bets accordingly.
(3) Capacity: at most max_open = 50 positions open simultaneously, enforced by a first-come queue.
(4) Settlement: each position settles at $t+H$ (with $H = 1$). The proxy profit is pnl_proxy = direction $\times z_{t+H}$.

The settlement proxy measures whether the z-score moved in the predicted direction and by how much, in z-score units. It is not a dollar profit metric and does not account for transaction fees or slippage.

3.6 Evaluation Metrics

We report both forecast quality and trading performance metrics:

- **R^2**: fraction of variance in the realized z_{t+H} explained by the model's predictions.
- **Directional accuracy (DirAcc)**: fraction of predictions where $\text{sign}(\hat{z}) = \text{sign}(z_{t+H})$.
- **Mean pnl_proxy**: average direction $\times z_{t+H}$ across settled trades.
- **Per-trade Sharpe**: mean(pnl_proxy)/std(pnl_proxy) on the filtered trade set (Rf = 0), used to select τ in Section 6.
- **Hourly portfolio Sharpe ratio**: $\text{mean}(\Delta\text{equity}_h)/\text{std}(\Delta\text{equity}_h)$ where hourly equity includes realized closed-trade profits plus mark-to-market valuation of open positions as direction$\times z_t$. Risk-free rate is zero because the z-unit proxy has no currency denomination and represents forecast quality rather than economic returns.

All metrics are computed with a matched **naive persistence baseline** ($\hat{z}_{t+H} \leftarrow z_t$) scored on the identical row set.

4 Experimental Setup

4.1 Data Collection

We collect live microstructure data from centralized cryptocurrency exchanges using a custom collector that polls 9 signal families (orderbook, trades, ticker, funding, open interest, withdrawal status, exchange status, spread matrices, OHLCV) via REST API at approximately 60-second target cadence across 23 assets and up to 6 exchanges (Binance, Bybit, OKX, Coinbase, Kraken, MEXC). Under full signal load, the observed cadence is approximately 110 seconds per snapshot. L2 orderbook imbalance, top-of-book sizes, and withdrawal/exchange status flags are not available from historical candle APIs and must be recorded live. Data collection protocol and signal schema are documented in the repository accompanying the public dataset release (URL withheld for double-blind review).

4.2 Asset Universe

The 23-asset universe was selected for high cross-exchange spread volatility and sufficient liquidity across multiple venues, requiring listing on at least 3 of the 6 target exchanges. See the repository for the full asset list.

4.3 Train / Test / Validation Split

All evaluation uses the same chronological partition; later sections refer to these sets by name without restating dates.

- **Training set.** Collection windows from June through mid-July 2026, strictly before the test cut. Approximately 2.9 million (coin, exchange-pair, snapshot) feature rows. Cross-sectional rows within a snapshot share market state and should not be treated as independent samples.
- **Test set.** July 25–28, 2026. Approximately 1.7 million feature rows collected directly from CEX API endpoints and stored so we can score the trained model (roughly a 63/37 train/test row split, or ≈1.7 : 1). The split uses no shuffling.
- **Validation set.** A sequential ≈72-hour live paper-trading campaign on August 4–7, 2026 (50,690 closed trades at the live gate; 1.01M scored predictions), run on real CEX feeds under the system's collection and decision latency. Capacity is max_open = 50. The paper confidence gate is τ=0.9 (Section 6); headline validation metrics use the post-hoc $|\hat{z}| \geq 0.9$ subset of this book unless noted.

4.4 Live Paper-Trading Protocol

For live validation, we run independent paper-trading campaigns that process incoming collector data in real time with no access to future snapshots. Capacity is max_open = 50. The paper confidence gate is τ=0.9 (Section 6). We validate on the August 4–7 ≈72-hour campaign using the same model and parameters as in training and testing.

An earlier short live session under the same model settings (H=1, w=300) supplies the capacity-matched mechanical baseline comparison in Section 5; we treat it as part of the validation evidence rather than a separate dated campaign.

4.5 Baseline Definitions

4.5.1 Naive persistence. On every row where the model generates a prediction, we also record the naive forecast $\hat{z}_{t+H} \leftarrow z_t$ (the current z-score predicts the future z-score unchanged). R^2 and directional accuracy are computed for both model and naive on the *identical* rows, ensuring that any performance difference is attributable to the model rather than to row selection.

4.5.2 Mechanical z-score baselines. We replay two mechanical trading rules on the validation signal panel under the same capacity constraints (max_open = 50), settlement proxy, and paper gate τ=0.9 as the model:

- **Persistence**: enter when $|z_t| \geq 0.9$, direction = sign(z_t).
- **Mean-reversion**: enter when $|z_t| \geq 0.9$, direction = $-$sign(z_t).

Scarce slots are filled by ranking $|\hat{z}|$ (model) or $|z_t|$ (mechanical). These represent the two canonical directions a rule-based z-score system can take: betting the spread continues (persistence) or reverts (classical pairs trading). Both settle at $t+1$ with pnl_proxy = direction × z_{t+1}, identical to the model's settlement.

4.5.3 Tree ensemble control (Random Forest). On the same out-of-sample **test** set used for LightGBM, we score a sklearn Random Forest regressor [1] trained on the same chronological feature matrix (a classical Random Forest baseline as a control within the tree family; Section 5). Primary live claims remain capacity-matched mechanical baselines on the validation panel.

4.6 Reproducibility

All training, evaluation, and baseline scripts are available in the code repository (the code repository (URL withheld for double-blind review)), and the dataset is released at the public dataset release (URL withheld for double-blind review).

5 Results

We evaluate the LightGBM z-score forecasting model on the test set and the validation campaign (Section 4.3). In each setting we report forecast quality metrics (R^2, directional accuracy) and trading performance metrics (mean profit proxy, hourly portfolio Sharpe ratio), always scoring a naive persistence baseline on the same rows as the model. The paper confidence gate is τ=0.9 (justified in Section 6).

5.1 Test-Set Evaluation

Table 1 reports forecast metrics on the full prediction set and on the paper-protocol filter $|\hat{z}| \geq 0.9$.

Table 1: Test-set evaluation. Filtered rows use the paper gate $|\hat{z}| \geq 0.9$.

Set	R^2	DirAcc
Test (all predictions)	0.133	62.8%
Test ($|\hat{z}| \geq 0.9$)	0.535	85.3%
Train (all predictions)	0.128	61.8%

Train and test R^2 are nearly identical on the unfiltered set (0.128 vs 0.133), indicating no overfitting (early stopping selected iteration 74 of 2,500). Under τ=0.9, test DirAcc rises to 85.3% and R^2 to 0.535. Validation τ=0.9 subsets (Section 6) remain consistent with this filtered test skill (DirAcc 86.7%, R^2 0.599), so both held-out regimes support the same selective-forecast story.

5.2 Tree Ensemble Control: LightGBM vs Random Forest

As a classical Random Forest baseline within the tree family, we train a Random Forest [1] on the same chronological feature matrix and score it on the **test** set alongside LightGBM and naive persistence $\hat{z} \leftarrow z_t$ (Table 2).[1] Predictions are highly correlated (corr ≈ 0.98), but Random Forest magnitudes are compressed by leaf averaging, whereas LightGBM's residual boosting keeps more mass in the prediction tails, so fewer Random Forest rows clear the absolute gate $|\hat{z}| \geq 0.9$. We therefore report n with every filtered metric: Random Forest's slightly higher own-gate DirAcc is a thinner slice, not a matched-row win. On LightGBM's $|\hat{z}| \geq 0.9$ rows, DirAcc ties at 85.2% while LightGBM retains the R^2 advantage (0.525 vs 0.489). We keep LightGBM as the reported model because, at the paper gate, it yields about twice as many test trades, higher R^2, and larger total `pnl_proxy` mass (mean × n).

Table 2: Test-set tree-family control on the shared LOGO feature matrix. Filtered rows use each model's own $|\hat{z}| \geq 0.9$ mask (except naive, which filters on $|z_t| \geq 0.9$). Bold marks the preferred LightGBM operating point on n and R^2.

Model	Filter	n	DirAcc	R^2	Mean pnl
LightGBM	all	1,680,426	62.8%	**0.133**	0.269
Random Forest	all	1,680,426	62.9%	0.126	0.267
Naive z_t	all	1,680,081	67.0%	−0.287	0.302
LightGBM	$|\hat{z}| \geq 0.9$	**66,513**	85.2%	**0.525**	1.154
Random Forest	$|\hat{z}| \geq 0.9$	33,038	86.0%	0.478	1.299
Naive z_t	$|z_t| \geq 0.9$	602,453	69.5%	−0.523	0.545

[1] Table 2 uses the shared LOGO feature cache (62 columns; six Coinbase volume lags pruned), so LightGBM τ=0.9 figures (85.2% / 0.525) differ slightly from the production booster in Table 1 (85.3% / 0.535).

5.3 Validation Campaign

Under the paper protocol $|\hat{z}|\geq 0.9$ applied to settled validation closes, the book has n=12,795 trades with DirAcc 86.7%, R^2 0.599, mean z-proxy +1.37, and per-trade Sharpe 1.08 (Table 4).

5.4 Baseline Comparison: Learned Forecast vs Mechanical Z-Score Rules

To isolate the value of the learned forecast from the confidence filter, we replay two mechanical baselines on the validation signal panel under matched capacity (max_open = 50) and the same magnitude gate τ=0.9. Both baselines enter when $|z_t| \geq 0.9$ but use the current z-score rather than a predicted future z-score to determine trade direction; scarce slots are filled by ranking $|z_t|$ (LightGBM ranks $|\hat{z}|$):

- **Persistence**: direction = $\text{sign}(z_t)$. Bets that the current z-score will continue into the next snapshot.
- **Mean-reversion**: direction = $-\text{sign}(z_t)$. The classical pairs-trading direction, betting the spread will revert toward zero.

Table 3 reports the capacity-matched baseline comparison, which is the primary evidence for the model's improvement over rule-based z-score thresholds.

Table 3: Capacity-matched baseline comparison on the validation panel at paper gate τ=0.9 (max_open = 50, ranked fill). Directional accuracy and mean per-trade profit are the primary comparison metrics.

Strategy	Direction	n closed	DirAcc	Mean PnL
LightGBM	$\text{sign}(\hat{z})$	2,026	**84.9%**	**+1.19**
Mech. persistence	$\text{sign}(z_t)$	8,550	69.5%	+0.73
Mech. mean-reversion	$-\text{sign}(z_t)$	8,550	30.5%	−0.73

As shown in Table 3, the learned policy improves directional accuracy by 15.4 percentage points and mean per-trade profit by 63% over mechanical persistence under the same τ=0.9 gate and capacity. Classical mean-reversion produces negative z-unit returns at this hold horizon. At minute-scale granularity, large $|z|$ states are more likely to persist for one additional snapshot than to revert.

Figure 2 and Figure 3 show capacity-matched directional accuracy and mean per-trade profit for all three policies on the validation panel.

5.5 Portfolio Risk-Adjusted Performance

Over the active hours of the validation τ=0.9 ranked-capacity baseline book, the model's hourly portfolio Sharpe ratio is 1.90 (mean hourly z-unit profit divided by its standard deviation, including mark-to-market). Capacity-matched mechanical persistence can reach about 2.62 here only because it closes far more trades ($n\approx$8,550 vs 2,026): hourly Sharpe rewards a thicker aggregated path, not better per-trade forecast skill—DirAcc and mean z-proxy still favor LightGBM (Table 3; Section 7). On the full ≈72-hour validation closes at τ=0.9, closed-only hourly Sharpe is 1.83.

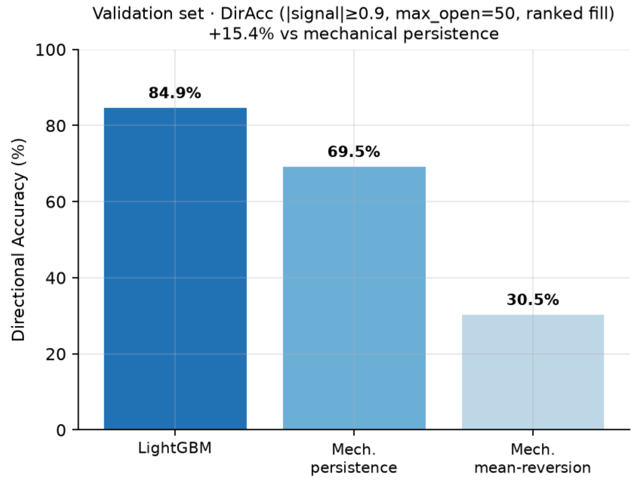

Figure 2: Validation set: capacity-matched directional accuracy at τ=0.9 (max_open = 50, ranked fill). The learned policy improves DirAcc by 15.4 pp over mechanical persistence.

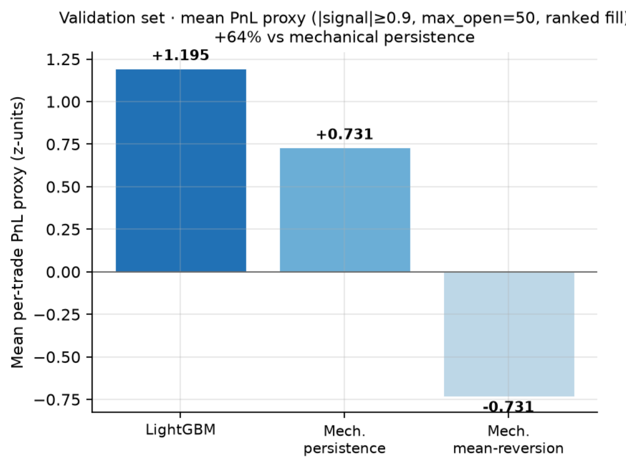

Figure 3: Validation set: capacity-matched mean per-trade PnL proxy at τ=0.9 (max_open = 50, ranked fill). The learned policy improves mean z-unit profit by 63% over mechanical persistence; mean-reversion is negative.

6 Model Tuning & Ablation Studies

6.1 Selecting Window Length and Confidence Threshold

Window w. Increasing w enriches $(\hat{\mu}_w, \hat{\sigma}_w)$ but forces longer warmup before a valid z exists and slows σ adaptation. Spreads-only LightGBM sweeps on the test set show skill rising into a plateau near $w\approx$560–720, with only modest gains past the production choice w=300 (≈+1 pp DirAcc and +0.017 R^2 at w=720 for ~2.4× lookback). Figure 4 makes the trade-off explicit under the paper gate $|\hat{z}| \geq 0.9$: past w=300, further increases mainly buy longer context for marginal filtered DirAcc. We therefore retain w=300.

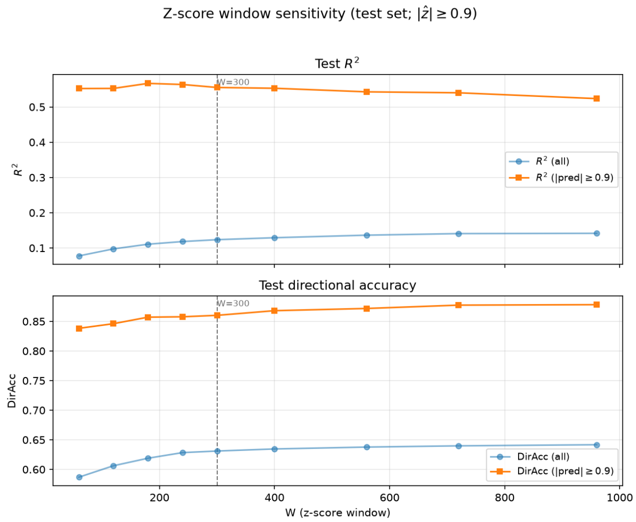

Figure 4: Z-score window w vs test R^2 and directional accuracy under $|\hat{z}| \geq 0.9$. Past w=300, additional lookback mainly increases required context while DirAcc gains flatten.

Threshold τ. Early live collection used a permissive gate τ=0.5 to maximize coverage under max_open=50. That choice is a starting point for selection, not the paper operating point. A post-hoc sweep on validation closes (w=300) shows that raising τ from 0.5 to 1.0 monotonically improves DirAcc, R^2, mean z-proxy, and per-trade Sharpe mean(pnl_proxy)/std(pnl_proxy) (Rf = 0), while cutting n (Table 4). **Per-trade** Sharpe first reaches ≥1 at τ=0.9: DirAcc 86.7%, R^2 0.599, mean z-proxy +1.37, Sharpe 1.08 on n=12,795 closes—still a large book relative to τ=1.0 (n=9,390), with a clear lift versus τ=0.5 (n=50,690; DirAcc 79.0%; Sharpe/trade 0.75). Neighboring points bound the choice: τ=0.75 remains below Sharpe 1, while τ=1.0 adds limited Sharpe for another ~27% coverage cut. We therefore fix τ=0.9 as the paper protocol; headline results use $|\hat{z}| \geq 0.9$ (including post-hoc filters on live books filled at a lower τ).

Table 4: Choosing τ on the validation set (w=300; post-hoc filter on live closes). Per-trade Sharpe uses Rf = 0.

τ	n	DirAcc	R^2	Mean	Sharpe/tr
0.5	50,690	79.0%	0.439	+0.84	0.75
0.75	20,866	84.5%	0.546	+1.17	0.95
0.9	12,795	86.7%	0.599	+1.37	1.08
1.0	9,390	88.0%	0.635	+1.51	1.17

6.2 Feature Importance

Figure 5 shows the top 10 of the model's 68 features ranked by LightGBM split gain.

The model mainly relies on z-score and spread lags. This semi-autoregressive structure makes sense given the problem context, wherein the previous states are strong indicators of future motion of the spread and z-scores. The coin and pair categorical variables are also included in the top 10, indicating that the model learns to differentiate between different coins on different exchanges. Finally, the microstructure features, namely trade data and L2 order book data, are also included in the top 10. Even though they are of smaller importance, these features provide some insight to the model.

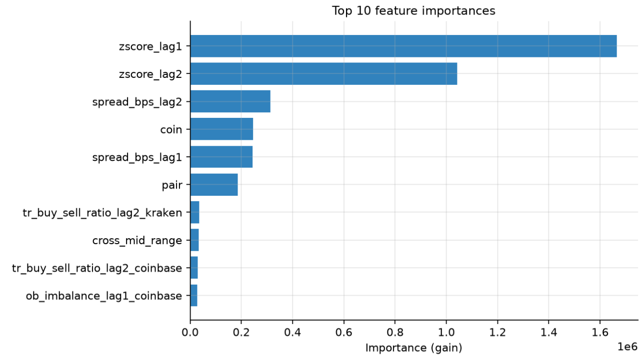

Figure 5: Top 10 features by LightGBM split gain.

Table 5: Engineered vs non-engineered microstructure features (test set, paper gate $|\hat{z}| \geq 0.9$). Identity (`coin`/`pair`) held fixed. Δ vs raw (non-engineered) columns: DirAcc_f +7.6 pp, R^2_f +0.523.

Variant	R^2_f	ΔR^2_f	DirAcc_f	ΔDirAcc_f
Raw + identity	0.012	—	77.8%	—
Engineered + identity	0.535	+0.523	85.4%	+7.6 pp

6.3 Ablation: Pre-Training Feature Engineering vs. Raw Market Features

The prediction target is $y_t = z_{t+1}$ (H=1). We ask whether *transforming* the same live microstructure—mid prices into spreads, rolling z-scores, lags, and short momentum/acceleration—improves filtered skill relative to training on raw live columns alone.

Engineered trajectory. From venue mids we form pairwise spreads S (bps) and rolling z-scores z, then construct lagged S and z, plus short momentum and acceleration of those series (9 columns). This is a pre-training transform of collected market data, not a change to what the collector observes.

Raw market features. Ticker mids and bid–ask widths, L2 imbalance, trade volumes, and cross-venue aggregates (51 surviving columns)—the same session's live signals without that trajectory transform.

Identity controls. `coin` and `pair` are held fixed on both sides of the comparison. They are not the engineering question: `coin` selects which asset's dynamics apply; `pair` selects which two venues the cross-exchange dislocation (and mean reversion) lives on.

We retrain LightGBM on the test set under fixed published `LGBM_PARAMS` and a shared boosting-round budget (49 rounds). Filtered metrics use $|\hat{z}| \geq 0.9$ (Table 4).

Result. The gain comes from transforming the initial microstructure, not from seeing different data. Compared to non-engineered microstructure features (raw ticker/OB/trade/cross columns with identity fixed), the mid→spread→ z trajectory improves filtered directional accuracy by +7.6 **pp** (77.8%→85.4%) and filtered R^2 by +0.523 (0.012→0.535). Under the paper gate, pre-training feature engineering on live mids is what moves next-snapshot z prediction, and the same lift carries through to higher mean `pnl_proxy` and per-trade Sharpe.

7 Discussion

7.1 Forecast Quality vs Economic Settlement

The paper's primary claim is forecast quality under a selective confidence gate: on the test set and the validation campaign, LightGBM improves directional accuracy and mean z-unit settlement over capacity-matched mechanical persistence (Section 5). The settlement proxy $\mathrm{pnl}_{\mathrm{proxy}} = \mathrm{direction} \times z_{t+H}$ is designed for that claim—it scores whether the standardized spread level moves as predicted—not as a dollar P&L.

When the same validation τ=0.9 ranked-capacity trades are rescored in basis points of realized spread change, policy ordering can invert (Table 6): z-unit wins for the learned policy do not automatically translate into positive mean gross bps. That gap is expected under the training objective (predict z_{t+H}, which is dominated by the slow-moving rolling mean) rather than instantaneous ΔS_t. We therefore report z-unit DirAcc, R^2, and mean `pnl_proxy` as the headline evidence, and treat fee-aware bps as a separate economic layer for future work.

Table 6: Validation τ=0.9 ranked-capacity entries scored in z-units and basis points of realized spread change, gross of fees. Units answer different questions: z-proxy for forecast quality, bps for raw spread change.

Policy	Mean z-proxy	Mean gross (bps)	Gross win rate
LightGBM	+1.195	−0.94	34.9%
Mech. persistence	+0.731	−3.97	9.9%
Mech. mean-reversion	−0.731	+3.97	76.0%

7.2 Portfolio Stability Complements Per-Trade Skill

Under the τ=0.9 ranked-capacity baseline book, capacity-matched mechanical persistence posts a higher hourly portfolio Sharpe (2.62) than LightGBM (1.90). That ranking does *not* overturn the model's advantage. Hourly Sharpe scores the path of *aggregated* hourly z-proxy, not forecast quality per fill: persistence admits about 4× as many closes (n≈8,550 vs 2,026) because $|z_t|\geq 0.9$ fires more often than $|\hat{z}|\geq 0.9$, so its hourly P&L series is thicker and can look stabler even though each trade is weaker. On the metrics that match the paper claim—per-trade DirAcc (84.9% vs 69.5%) and mean z-proxy (+1.20 vs +0.73) under the same gate and capacity—LightGBM remains clearly ahead. We therefore treat hourly Sharpe as a secondary stability check and rank policies by selective forecast skill.

8 Conclusion

We presented a framework for forecasting same-asset cross-exchange cryptocurrency spread dynamics under live multi-venue microstructure, using a confidence-gated tabular tree baseline rather than the DNN/LSTM architectures common in prior supervised spread work. Optimizing the rolling z-score window w and confidence gate τ yields an hourly Sharpe ratio of 1.90 under the paper protocol. On the live validation campaign, the LightGBM policy at $|\hat{z}| \geq 0.9$ achieves directional accuracy of 86.7% and R^2 of 0.599 on settled closes (n=12,795), improving per-trade win rate by 15.4 percentage points versus capacity-matched mechanical z-score persistence. The confidence filter lifts R^2 by selecting high-signal environments; on those rows, much of the incremental value is in capturing spread *magnitude*, not only direction. Engineered lag and momentum features explain a large share of predictive power, with a smaller but complementary contribution from real-time microstructure collected through exchange APIs.

As a classical Random Forest baseline—and a control that is still uncommon in this cross-exchange microstructure setting—we also score Random Forest on the same out-of-sample test set used for LightGBM. Predictions agree strongly (corr ≈ 0.98), confirming that ranking skill is not unique to gradient boosting, while LightGBM remains the reported model for higher R^2 and roughly twice as many trades at the absolute $|\hat{z}| \geq 0.9$ gate.

8.1 Future Work

In future, incorporating more realistic fee structures would bridge the gap between z-unit performance and economic profitability. The released dataset enables the community to benchmark alternative architectures—including deeper sequence models and stronger tabular peers—on the same cross-exchange microstructure panel.

Ethics and Privacy Statement

This study uses only publicly available aggregate market data (quotes, orderbook depth, anonymous trade prints, funding rates) from public REST endpoints of centralized cryptocurrency exchanges. No personal data or account identifiers are present; no human subjects were involved. Data collection respected each venue's published rate limits; no authenticated or access-restricted endpoints were used.

All trading results in this paper are from simulated paper trading—no live orders were submitted and no capital was deployed. Reported performance uses z-score forecast-quality proxies rather than dollar P&L. The accompanying dataset release contains only the public market observations described above, enabling reproducible research on cross-exchange microstructure without exposing private account information.

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